

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 23

Team Members :

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1. Problem Title : Asset Tracking System

2. Problem Description :

In many organizations, assets are still managed manually, which often leads to problems such as misplacement, duplication of records, and inefficient usage. It becomes difficult to keep track of which asset is assigned to whom, where it is located, and how it is being used over time. As a result, maintaining accurate records and ensuring proper utilization becomes challenging.

To overcome these issues, an Asset Tracking System is proposed. This system helps in digitally registering assets, assigning them to employees, and monitoring their movement and usage. By maintaining all records in a structured way, the system improves transparency, reduces errors, and ensures better management of organizational assets.

3. Identified Entities (Classes)

Class Name	Description
Asset	This class represents all the assets available in the organization, such as equipment or devices, which need to be tracked and managed properly.
Employee	This class represents the employees who use the assets. It helps in keeping track of which employee is assigned which asset.
Storekeeper	This class represents the person responsible for handling and maintaining the assets, including assigning and receiving them.
Admin	This class represents the administrator who manages the overall system, monitors asset usage, and ensures everything runs smoothly.

AssetTransaction	This class is used to record all asset-related activities such as assignment, return, or transfer, helping in tracking asset movement.
Report	This class represents the reports generated by the system, which provide details about asset usage, allocation, and overall status.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Asset	assetId, assetName, assetType, status, location	addAsset(), updateAsset(), checkAvailability()
Employee	employeeId, name, department	requestAsset(), returnAsset()
Storekeeper	storekeeperId, name	assignAsset(), assignAsset(), maintainAsset()
Admin	adminId, name	monitorAssets(), generateReports(), manageUsers()
AssetTransaction	transactionId, assetId, employeeId, date, status	recordTransaction(), updateTransaction()
Report	reportId, reportType, generatedDate	generateReport(), viewReport()